Income Heterogeneity of Immigrants and House Price Dynamics

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Abstract

We investigate how high- and low-income immigrants affect house prices in neighborhoods in the Netherlands. To circumvent the endogeneity of immigrant location choice, we use a shift-share instrumental variable strategy, exploiting past settlement patterns of immigrants. We use individual administrative data linked to residential property values and sale prices between 2006 and 2022. Results show an average positive effect of 0.32 percentage points from an inflow of high-income immigrants on house price growth in the short run, while an increase in low-income immigrants decreases year-on-year growth of house values by 0.14 percentage points. The changes are mainly driven by postcodes in less urban municipalities. At an average change in the immigrant share (0.24%), property prices grow around 1 percentage point less for all types of immigrants. This medium-turn effect is potentially mediated by native outflow from areas that experience an increase in the immigrant share.

Keywords: Housing prices, immigration, urban areas
JEL Codes: J61, R23, R31

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Results in this paper are based on non-public microdata from Statistics Netherlands (CBS). In principle, part of these microdata is accessible for statistical and scientific research. For further information: microdata@cbs.nl.

1 Introduction

Mobility of individuals across borders and regions has been increasing worldwide over the last decades (OECD (2024)). Understanding the economic effects of immigration has been of interest for academic research and policy alike. Immigration has also become a very salient topic in politics, with concerns about labor markets, housing markets and integration of the foreign-born population. A large body of literature examines labor market effects of immigration and generally finds moderate adverse effects on employment and wages of natives (see an overview by Kerr and Kerr (2011) or Dustmann, Schönberg, and Stuhler (2016)). These effects often depend on the education and skill levels of both natives and immigrants (e.g. Card (2001)). A recent and growing literature has expanded to investigate the effect of immigration on the housing market. The housing market offers a different lens than the labor market, as any changes may affect a broader segment of the population, including non-working households such as retirees or students. An inflow of immigrants to a neighborhood can have effects on both house prices and attractiveness of the neighborhood. A price effect from immigration will directly affect household budget as households typically spend a substantial fraction of disposable income on housing.[1] Another closely linked dimension is attractiveness of an area and to what extent an inflow of immigrants leads to segregation and potentially out-mobility of local residents (see a discussion by Boje-Kovacs et al. (2024)). Existing evidence on how immigration affects house prices, rents, and native residential mobility remains mixed across countries and contexts, as reviewed in Cochrane and Poot (2021) and Larkin et al. (2019). In Section 2, we discuss more in detail how methodological choices — such as the spatial scale of analysis, the time horizon, and the specific population groups considered — may account for these divergent findings. Notably, most studies consider the general population of immigrants (foreign-born). We further distinguish immigrants by income classes, into high and low-income immigrants.

In this paper, we investigate how immigrants with heterogeneous income levels affect local house prices in the Netherlands. The main contribution of this paper is to distinguish immigrants by income classes, similar to labor market studies that distinguish immigrants by skill groups. Income is relevant for immigrant's purchasing power, but may also mediate native's attitudes to immigrants and the relative attractiveness of the neighborhood, if local residents'

[1] Housing expenditure of disposable income is around 23% in the Netherlands (Eurostat (2023)).

preference for co-location varies with immigrants' income. The Netherlands is a suitable setting for this study, for two main reasons. Firstly, administrative data is available to link population, residential location and immigrant settlement across small geographical units. Secondly, the Netherlands receives immigrants across a wide range of the income distribution, providing variation across the immigrant income distribution. Factors attracting high-skilled immigrants in particular include tax benefits ("30% rule") and a simplified visa procedure for skilled immigrants (*Kennismigrantenvisa*).[2]

Two main mechanisms are likely to drive the impact of immigrants on the housing market, as discussed in Sanchis-Guarner (2023). Firstly, an inflow of immigrants increases demand for housing and may induce competition in the housing market, as supply of housing is fixed in the short run. Assuming an upward sloping supply curve of houses, a demand increase mechanically increases house prices. Secondly, immigrants may affect the (perceived) amenities and attractiveness of the area they settle in. This may result in selective (out-) mobility of the resident population from a neighborhood, which may decrease house prices (see a discussion in Accetturo et al. (2014), Sá (2015) and Boje-Kovacs et al. (2024)). The direction of the effect of immigration on house prices is ex-ante ambiguous and may depend on the type of immigrants. Distinguishing immigrants by income levels is relevant for both of the above mentioned mechanisms. The (positive) demand effect may be more pronounced for an inflow of high-income immigrants as they may have more financial leverage to overbid when buying a house or renting. In such a scenario, overbidding increases transaction prices, and might deter individuals with lower willingness or ability to pay. The second aspect of (native) out-mobility depends on whether natives' willingness to co-locate with immigrants varies with immigrant income. Recent literature on residential sorting finds that households tend to prefer neighborhoods with a higher share of high-income households (for example Duijn and Rouwendal (2021)). Therefore, immigrant's income level may mediate residents' tendency to move away from immigrant dense neighborhoods.

The literature on the housing market effects of immigrants finds mixed results in terms of direction and magnitude, as reviewed in Cochrane and Poot (2021) and Larkin et al. (2019). Some

[2]The "30% rule" offers a tax reduction to skilled immigrants for a limited period in time. Timm, Giuliodori, and Muller (2025) shows this tax benefit attracts immigrants with mid-range incomes. The *Kennismigrantenvisa* provides simplified visa procedure for firms to hire non-EU skilled immigrants.

of these mixed findings are due to differences in the spatial unit and the time frame. The size of the spatial unit (e.g. neighborhood versus provinces) affects to what extent estimates pick up any adjustments from residential mobility. On the other hand, the choice of time frame matters for the elasticity of housing supply in the short run or long run, but also for resident's relocation decision. Notably, most studies consider the general population of immigrants (foreign-born). In this paper, we estimate the effect of a change in the share of immigrants distinguishing by income classes. This approach further nuances how different types of immigrants affect house price dynamics. We will review and discuss related literature in detail below in Section 2. Our contribution to the existing literature is twofold. First, we explicitly account for immigrant's income level and employment. This allows to further differentiate how different characteristics of immigrants affect house prices and amenities. This approach further nuances findings from preceding literature, where immigrants are commonly treated as a homogeneous group. Second, we use granular administrative data, representative of the whole Dutch population and property values. This enables us to link immigrants with their residential location and income and employment information. While preceding literature examines the impact of immigration on house prices in the US, England, Italy and Spain, we add such a study for the Netherlands.

Estimating a causal effect of immigration requires dealing with the endogeneity of immigrant location choice, as residential location choice is unlikely to be random. If immigrant location choice is driven by factors that affect also the change in house prices, simple linear regression is biased. To deal with this issue, we use a shift-share instrument that predicts the contemporaneous change in the immigrant stock in a neighborhood by interacting the historical settlement shares of immigrants by region of origin with current national-level inflows from the same region of origin.This type of instrument is commonly used in the literature of economics effect of immigration and is based on a network effect that immigrants tend to co-locate with immigrants of the same origin.[3] We provide in Section 4 further motivation and descriptive evidence that immigrants indeed cluster by country of origin. We categorize immigrants by their position relative to quartiles of the annual municipality-level income distribution. We use the quartiles as a proxy for the immigrant's position relative to the wider (municipality-level)

[3]For example Card (2001) for labor market effects, Goldsmith-Pinkham, Sorkin, and Swift (2020) and Jaeger, Ruist, and Stuhler (2018) provide an in-depth methodological review and examples.

income distribution.[4] We estimate the contemporaneous effect of a change in the share of immigrants relative to the population on the change in property value in a neighborhood. Our analysis combines population-wide administrative data from Statistics Netherlands (CBS) with transaction records from the Land Registry (Kadaster), covering the period 2006 to 2022. This detailed data enables us to track demographic composition and housing market dynamics at a disaggregated level.

We find that, on average, an increase in the share of immigrants is associated with a small positive change in local property values. However, we find that the effect of immigration on local property values is heterogeneous by immigrant income, and that this heterogeneity is masked in pooled estimates. A one percentage point increase in the share of middle-low income immigrants — the most precisely estimated effect — raises annual property value growth by roughly 0.27 percentage points at the mean inflow, or approximately one-sixth of mean annual appreciation. Middle-high income immigrants generate a similarly positive effect of 0.44 percentage points at the mean. By contrast, an equivalent inflow of low-income immigrants reduces annual property value growth by 0.14 percentage points, an effect that grows in magnitude once local demand controls are included, suggesting that omitted factors if anything attenuate the negative effect. High-income immigrants generate a positive but imprecisely estimated effect.These effects are robust to modifications of the share and the shift component of the instrumental variable and to the inclusion of additional controls for population, housing stock and unemployment rate. Turning to heterogeneity by area, we find that these effects are driven by postcodes in a set of "less urban" municipalities (less than 100,000 inhabitants in 2021). We show descriptive statistics that the immigrant share is on average lower in less urban neighborhoods and we argue that therefore a change in immigrant inflow may be more salient. We present additional indicators from the *World Value Survey* showing that attitudes towards immigrants vary between cities and less urban areas. From a small but representative survey, it seems that respondents in larger towns are more open to having immigrants as neighbors and seem to be less concerned about labor market competition. In the medium term, a three-year horizon, all effects reverse sign, consistent with a native displacement mechanism whereby an initial demand-driven price

[4]We will refer to the top quartile as "high income", the second quartile as "midhigh income", the third quartile as "midlow income" and the bottom quartile as "low income". Non-working immigrants are assigned income zero.

increase is followed by incumbent outflow that erodes the local demand base.

The paper proceeds as follows. Section 2 reviews and discusses related literature. Section 3 describes the data and presents descriptive statistics of immigrants and neighborhoods. Section 4 explains the empirical framework of the shift-share instrumental variable. Section 5 presents the results, a robustness exercise and heterogeneous effects by area. Finally, Section 6 provides a discussion and concludes.

2 Related literature

Our study relates to a strand of literature examining the relation between international migration and housing and rental markets. Two meta studies by Cochrane and Poot (2021) and Larkin et al. (2019) conclude that on average, the effect of immigration inflows on rental and housing prices is positive, however with large variation across studies. The direction and size of the estimated effect of immigration on house prices substantially vary depending on the regional definition and time frame, with some studies finding negative effects (Saiz and Wachter (2011), Accetturo et al. (2014), Sá (2015), Boje-Kovacs et al. (2024)). We summarize the results of the main studies in Appendix Table A.1, and discuss here how they structurally differ.

Firstly, the choice of the area matters. Some studies focus on broader regional level, for example provinces, while the most disaggregated level studied is the building level.[5] Most studies conducted on a larger spatial unit tend to find positive effects while most studies measuring outcomes on a more granular level find negative effects of immigration inflow on rental and housing prices (see Appendix Table A.1). Notably, Accetturo et al. (2014) contrast the effect at different levels of spatial aggregation and find an increase in house prices at city level but a decrease at district level within cities in Italy. The estimated effect of immigration on house prices may vary by geographical scope for two main reasons: larger geographical areas may mask mobility within the region (i.e. from one neighborhood to another within the same city, see the discussion in Accetturo et al. (2014)). In this case, direct demand for housing by immigrant

[5]From larger to smaller areas: Gonzalez and Ortega (2013) for provinces in Spain, Saiz (2007) and Mussa, Nwaogu, and Pozo (2017) for metropolitan statistical area (MSA) in the USA, Saiz and Wachter (2011) using census tracks in the USA, Sá (2015) for local authorities in the UK, Accetturo et al. (2014) for cities and districts in Italy, Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019) for neighborhoods in Spain and Boje-Kovacs et al. (2024) for neighborhoods and building units in Denmark.

inflow may be the main driver of the effect (see discussion by Larkin et al. (2019)), implying an increase in prices for a net positive inflow of immigrants. Secondly, in a smaller area incumbent residents may have more contact with potential newcomers. Therefore, interaction may be more salient in smaller or more densely populated areas (see the discussion by Boje-Kovacs et al. (2024)). Depending on preferences of incumbent residents, this may result in (selective) out-mobility, as shown in Sá (2015). Our study analyzes the effect of immigration on house prices using data disaggregated at 4-digit postcode level. This level of granularity is relatively small compared to the existing literature.

Secondly, the time frame of the analysis matters. Housing supply is likely inelastic in the short run as it is potentially restricted by building permits and construction time. Secondly, relocation decision may not be immediate. As highlighted by Saiz (2003), adjustment of prices through arbitrage may happen in the medium to long run because of mobility and information frictions. Moreover, Akbari and Aydede (2012) find relatively small effects in the overall housing market in Canada and only from immigrants that arrived more than 10 years before. They reconcile that effect by arguing that most immigrants rent first and that (out-) mobility decisions of incumbent residents may be lagged in time. In this paper, we show both contemporaneous (year-on-year) and medium term (3-year difference) effects. For a general inflow of immigrants, we find a positive effect in the short run and a negative effect in the medium run.

Both aspects of area choice and the choice of time frame relate to the broader question whether immigrant inflow triggers native outflow. This is relevant for the housing market and also has been studied in the context of labor market effects, but also presents mixed findings in the literature. Several contributions highlight a role of immigration for native residential mobility. Examining a large immigration shock in Spain, Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019) find a strong response of natives to move out of narrowly defined neighborhoods that experience an inflow of immigrants. For the UK, Sá (2015) finds that high-income households leave from neighborhoods with a large increase of immigration inflow. This leads to a decrease in the average income in the neighborhood and contributes to the negative effect on housing prices. Hatton and Tani (2005) also find displacement of natives in response to an inflow of immigrants, although with varying magnitude across the UK. Mocetti and Porello (2010) and Card and DiNardo (2000) consider the skill composition of natives for Italy and the US respectively. The former paper finds complementarity of immigrants with highly-educated

natives and outflow of low-educated natives, while the latter finds small increases of natives in accordance with the immigrant's skill group.

The main argument for out-mobility of local (native) residents upon an inflow of immigrants is the unwillingness to co-locate with immigrants. In a meta study, Larkin et al. (2019) find that the effect of immigration on housing prices is more likely to be positive in places where locals are more accepting of immigrants as measured by national responses to the World Value Survey (WVS). Larkin et al. (2019) argue that in those places, increased demand for housing is the main driver of the effect as natives' willingness to pay to co-reside with immigrants is less negatively affected. Additionally, in the case of positive attitudes towards immigrants native out-mobility is a less salient channel. In this paper, we present survey responses from the World Value Survey and show that the natives' acceptance of immigrants also varies by location (less accepting in rural areas, more open in urban areas). While the answers to the World Value Survey are only indicative, they are informative to our heterogeneity exercise where we find a stronger response to immigrants in less urban areas. Unfortunately, the questions from the World Value Survey do not explicitly include immigrants' income. How attitudes to immigrants vary with immigrants' income is a topic for future study.

The above mentioned papers mostly consider immigrants as a homogeneous group, namely foreign-born adults (see Appendix Table A.1). In this paper, we differentiate immigrants by income and classify immigrants by the quartiles of the municipality-level income distribution. While income may be biased due to discrimination, wage bargaining or down skilling, it does serve as a proxy for an individual's position in the income distribution. Additionally, income tends to be correlated with the level of education and productivity, two important factors that may contribute to making a neighborhood more attractive.

Beyond a general definition of immigrants, there is a strand of literature that considers specific types or events of immigrants. Notably, this allows to use a difference-in-difference or regression discontinuity design exploiting exogenous events. Saiz (2003) investigates the effect of a large inflow of Cuban immigrants in Florida after the immigration shock from the "Mariel boatlift", finding a large persistent increase of 7-8% of rent prices in Miami. On a very local level, Dröes and Koster (2023) consider granular corridors around refugee settlement centers in the Netherlands and find a very locally concentrated decrease in housing prices in places close to pathways of refugees. Gautier, Siegmann, and Van Vuuren (2009) consider how house prices

in Amsterdam neighborhoods with a large share of Muslims ($> 25\%$) react to a fundamentalist-Islamic terrorist attack. They find a 2.4% decrease in house prices in neighborhoods with a large Muslim population, relative to comparable neighborhoods.

The most common explanation from the literature on how immigrants affect house prices are twofold. Firstly the increase in demand combined with a short-run inelastic housing supply and secondly the outflow of natives in response to migration. In the short run, increased demand from immigrants may drive up rental prices and potentially house prices. Considering the demand for rental units, Saiz (2003) finds an increase in rents of low-quality units typically inhabited by immigrants, but not for high-quality rental units, which can be reconciled with an increase in demand in a particular segment of the market.

Our contribution to the existing literature is twofold. First, we differentiate for immigrant's income level and employment. Most other studies consider immigrants as a homogeneous group of foreign-born individuals. The exception is Sá (2015), who also considers heterogeneity by immigrants' level of education. They find that the overall negative effect is mainly driven by regions which a high share of low educated immigrants. Sá (2015) divides the units of observation (local authorities) the share of immigrants with low education. In our study, we differentiate by immigrant income instead of differentiating by regional characteristics. Second, with the level of detail available in the administrative data, our study adds to the few papers inspecting relatively small spatial units, departing from most studies that use provinces (Gonzalez and Ortega (2013)) or commuting areas (Saiz and Wachter (2011), Mussa, Nwaogu, and Pozo (2017)). In terms of the choice of spatial unit, our study is closely related to Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019) who study neighborhoods in Spain and Boje-Kovacs et al. (2024) who study the impact of non-Western refugees in neighborhoods and buildings in Denmark.

3 Data

We combine individual level administrative data from Statistics Netherlands (CBS) with residential property records and the Land Registry (Kadaster) for the years 1995 - 2022. This allows us to examine how house prices respond to demographic changes within narrowly defined neighborhoods. In the following, we describe the data and main variables.

In the main analysis, we will examine how demographic changes within a neighborhood,

specifically the share of immigrants, affect house prices. We conduct the analysis on the neighborhood level, which we define as the 4-digit postcode level.[6] Using the neighborhood definition of a 4-digit postcode allows us to link individual-level data with a rich set of neighborhood-level data (*Kerncijfers Wijken en Buurten* (KWB)). This contains information on inhabitants, houses and amenities, such as population density, demographic structure, employment and income information, housing stock and valuation, housing stock composition (rental, owner-occupied, and corporately owned), distance from general practitioner, supermarket, and schools. The average size of 4-digit postcode territory is 2.5km^2 (Veldhuizen, Stronks, and Kunst (2013)). We argue that this is an appropriate size such that inhabitants can relate to their surroundings and may respond to demographic changes in this narrowly defined area. We will show that there is substantial variation across postcodes in descriptive statistics.

We select a sample of 1,795 postcodes from a total of around 4,000 postcodes, as shown in Table 1. Firstly, we take a balanced panel of postcodes that exist throughout 1995 - 2022. Secondly, we select postcodes with positive values of population and immigrants for all years. Thirdly, we exclude postcodes with year-on-year population changes exceeding 100% to avoid outliers. This step removes a small number of postcodes (50-80). Finally, we choose a cut-off for a minimum size of population in 1995, to avoid sparsely populated postcodes. We impose a minimum population threshold in 1995, using the 30th percentile of the population distribution across postcodes (539 inhabitants). We show robustness to alternative cutoffs in Section 5. We retain a sample of 1,795 postcodes, with an average stock of inhabitants of 5,850, where 1,110 are immigrants, and 4,638 dwellings. We cover at least 80% of population, immigrants and housing stock relative to the full set of postcodes.

Our analysis examines the impact of a change in (migrant) demographic composition on house prices. As highlighted in the literature (for example Boje-Kovacs et al. (2024)), both native's preferences to co-locate with immigrants as well as demand for amenities by different sociodemographic groups may play a role in the attractiveness of a neighborhood and hence housing demand. We draw information on inhabitants and immigrants from administrative registry data, available since 1995. We define immigrants as non-Dutch individuals settling from abroad in a Dutch municipality at an age of at least 18 years. We focus on a somewhat narrow

[6] In the following, we use "neighborhood" and "4-digit postcode" interchangeably. The most detailed postcode level in the Netherlands has 6 digits.

Table 1: Number of postcodes by selection step

Filtering Step	N postcodes	Population stock		Migrant stock		Housing Stock	
		Mean	Total (1995)	Mean	Total (1995)	Mean	Total (1995)
All postcodes	4,061	3,291	12,189,241	560	1,728,899	2,668	10,839,566
> 0 inhabitants	2,980	3,912	10,664,248	704	1,608,790	3,165	9,437,288
> 0 migrants	2,802	4,035	10,391,263	735	1,587,196	3,263	9,150,348
Growth rate inhabitants	2,751	4,065	10,290,622	743	1,577,069	3,276	9,020,298
Growth rate migrants	2,671	4,169	10,251,087	764	1,574,609	3,354	8,965,230
30th pct population (1995)	1,795	5,850	9,763,284	1,110	1,541,418	4,638	8,330,856

Note: Each line represents a successive sample selection. Population and number of immigrants both refer to individuals age ≥ 18. Housing stock is defined as the number of dwellings with a registered address. Postcode refers to a 4-digit postcode. 'Mean' refers to the mean variables across postcodes.
Source: *CBS Microdata*

definition of first-generation adult immigrants as opposed to more established second-generation immigrant, as we are interested in the impact of demographic changes within a neighborhood.[7] Moreover, we classify immigrants into different income groups, using the quartiles of the annual municipality-level income distribution as reference. This gives an indication whether a working immigrant earns high or low income relative to their municipality of residence. Considering this municipality annual average income, instead of the national one, accounts for local economic conditions and regional disparities. This gives a better assessment of the income level of immigrants with respect to the local area of residence. We will provide descriptive statistics when discussing the empirical strategy.

The key outcome is the change in house prices at the neighborhood level. We will in turn describe two measures, property value and house price sale index. Firstly, we use a measure of property value aggregated by neighborhood. The property value, or WOZ (*Waardering onroerende zaken*), is estimated annually for every property by the municipalities and is used to determine property taxation. It is assessed considering sale prices of comparable properties, square meters, location and other characteristics of properties sold in the surrounding area in years close to the valuation. The WOZ value for a given property typically refers to the sale

[7] "First generation immigrant" typically refers to a person born abroad that immigrated to a country. "Second generation immigrant" is defined as a person born in a given country, but with parents that are born abroad. In the administrative data we observe both the country of birth and the parents' country of birth, but not the migration motive (e.g. work or family reunion).

prices of the previous one to three years. Hence, we will take the lead of the average WOZ value per neighborhood. Secondly, we construct the Case-Shiller repeat sales index from the transaction prices of existing owner-occupied dwellings, recorded by the Land Registry (Kadaster).[8] We calculate the change in the transaction price for every house sold at least twice in the period 1995 - 2023, setting the base year to 2015. The construction of the index following Case and Shiller (1987) is described in Appendix Section D. This index reflects the change in sale prices of sold properties per postcode. The index is based on 56,500 to 126,500 annual transactions in our sample, where on average 40-50 properties are sold per neighborhood and year.

The main conceptual differences between the repeat sales index and the WOZ value are the following. Firstly, the repeat sales index is based on actual transactions only, while the WOZ value is estimated for every property and year. The repeat sales price index is therefore representative for a neighborhood to the extent to which house transactions are representative for a neighborhood in terms of appreciation and turnaround times. This may induce some selection issue in the repeat sales index, for example if prospective sellers postpone transactions in a moment of unfavorable macroeconomic conditions. Secondly, while we have property-level sale prices to calculate the repeat sales index, our dataset contains the WOZ value on the neighborhood level only. This entails that the average WOZ per neighborhood treats a neighborhood as a homogeneous unit. Thirdly, the time reference differs. The WOZ value for every property is based on sale prices of observationally similar properties, in the last one to three years. It typically lags behind the sale price index, which tends to pick up changes in transaction prices more quickly, if properties are sold.

Descriptive statistics

In the following, we briefly describe characteristics of the sample. Firstly, we show that levels and growth of population and houses vary across postcodes. Secondly, we describe characteristics of immigrants and immigration spells. Lastly, we describe developments of house prices.

The sample contains 1,795 4-digit postcodes from 346 municipalities with a total around 9.76 million individuals and 8.33 million housing units (in 1995). Table 2 summarizes descriptive

[8]The Land Registry records transactions of existing houses purchased by private individuals. This excludes newly built properties and transactions involving housing corporations.

Table 2: Descriptive statistics

	Years	25th Pct.	Mean	Median	75th Pct.	N
Avg. WOZ value by PC4 (x 1,000)	2006 - 2022	178.0	239.1	223.3	277.7	1,782
Δ Avg. WOZ by PC4 (%)	2006 - 2022	-1.8	2.9	2.6	7.0	1,779
Repeat Sales Index	2006 - 2022	1	1.12	1.08	1.19	1,272
Δ Repeat Sales Index	2006 - 2022	-3.2	2.1	2.0	7.1	1,272
Housing Stock	2006 - 2022	2,638	4,638	4,295	6,176	1,794
Δ Housing Stock (%)	2006 - 2022	0.00	-0.02	0.00	0.00	1,794
Share of rental units (%)	2006 - 2022	29	43	39	54	1,531
Share of owner-occupied units (%)	2006 - 2022	45	56	60	70	1,531
Population:						
Population age $\geq$ 18	2006 - 2022	3,442	5,850	5,450	7,808	1,794
Δ Population (%)	2006 - 2022	-0.33	0.52	0.32	1.09	1,794
Immigrants age > 18	2006 - 2022	379	1,110	769	1,380	1,794
Δ Migrants (%)	2006 - 2022	-0.62	1.57	1.20	3.32	1,794
Migrant share (%)	2006 - 2022	9	17	14	21	1,794
Avg. household size	2006 - 2022	2.0	2.2	2.2	2.4	1,782
Income:						
Avg. income per worker (x 1,000)	2009 - 2022	25.3	29.7	29.2	33.2	1,384
Share of low-income population (%)	2009 - 2022	36.5	40.7	40.3	44.4	1,391
Share of high-income population (%)	2009 - 2022	14.2	19.5	18.6	23.4	1,389

Note: Descriptive statistics across 4-digit postcodes and years. Δ denotes the year-on-year percentage change. Share of rental and owner-occupied units is relative to the stock of housing. Immigrant share, low-income and high-income share are relative to the population on Jan 1st in a given year. Share of low-income (high-income) population is defined as the number of households that belong to the bottom 40% (top 20%) of income in the entire country. Both share of low-income and high-income population are available from the KWB dataset and only indicated if there are at least 100 households by neighborhood.
Source: *CBS Microdata*, *Kerncijfers Wijken en Buurten (KWB)*

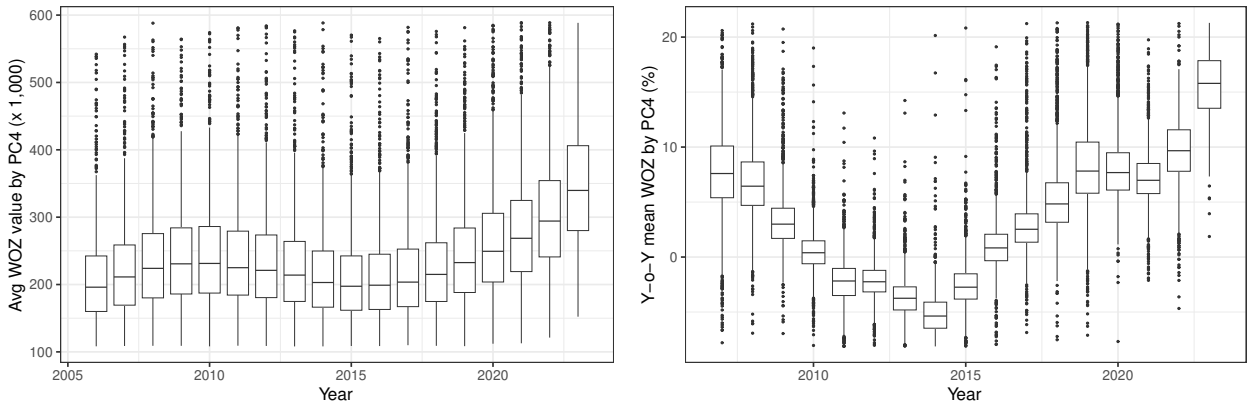
statistics across neighborhoods. The median postcode in the sample encompasses on average 5,450 inhabitants aged $\geq$ 18 of which 14% are immigrants. Across years and postcodes, the immigrant share varies substantially with an interquartile range between 9% (25th percentile) and 21% (75th percentile). We will use this variation of immigrants by neighborhood in our empirical strategy later on. The median housing stock per neighborhood is 4,295.

The majority of immigrants in this sample stay for a medium to long term in the Netherlands, making an impact on the neighborhoods and housing market plausible. Approximately 35% of

immigrants that arrived between 1995 and 2015 are still registered in the country by 2022 (see Appendix Chart E.3). Conditional on completing a immigration spell, about 75% stay for at least one year, with a median duration of approximately 2 years (see Appendix Table E.3). While being in the Netherlands, the median immigrant has 2-3 housing spells, where - conditional on completing a housing spell - 75% of individuals stay for at least one year at an address (see Appendix Table E.3). Therefore, short-term immigration from seasonal workers or high residential mobility is limited for the population in this sample. This strengthens the notion that immigrants reside for a few years in a neighborhood and therefore may impact the attractiveness of a neighborhood through demographic composition.

Figure 1 shows the level and growth of the WOZ value across neighborhoods over time. As shown in both Table 2 and in Figure 1, the WOZ value increases on average by 2.9% across all years and neighborhoods. Neighborhoods exhibit considerable differences in both the level and growth of the WOZ value as reflected by the spread across years. Note that the overall property value appreciation also varies substantially across years, with the years 2011 - 2015 marked by depreciation.

Figure 1: Levels and year-on-year (%) change in WOZ value across postcodes



Note: Levels of property WOZ value (LHS) and year-on-year growth of WOZ value (RHS) across years and postcodes. For every year, the boxplot shows the distribution of WOZ values across neighborhoods. Per year, the box shows the interquartile range (25th to 75th Percentile) and the line reflects the median. The whiskers reflect the range of values within median $\pm 1.5\times$ interquartile range. Dots reflect data points outside of the whiskers. Sample of 1,795 postcodes as outlined in Table 1.
Source: *CBS Microdata, Kerncijfers Wijken en Buurten (KWB)*

We show a similar figure for the repeat sales index in the Appendix Figure E.1. The repeat

sales index displays a similar trend as the WOZ value, with a drop in sale prices in 2011 - 2015 and a strong appreciation after. The correlation between the year-on-year change of (national level) WOZ value and repeat sales index is 0.86. From visual inspection, it seems that the repeat sales index decreases slightly less than the WOZ value in 2011 - 2015. This could be moderated by a selection in property transactions, if potential sellers postpone a transaction in a phase of low house values. Additionally, we show a comparison of country-level aggregate indices of the WOZ value and the repeat sales index in Appendix Figure E.2. While both measures exhibit some level differences, especially in the years 2011 - 2015, they co-move as shown by the year-on-year change of both indices. This underlines that the (lead of) WOZ value and the repeat sales index pick up a similar year-on-year variation and are appropriate measures of house prices developments in the Netherlands.

From correlations across postcodes and over time, we observe that the average WOZ value had a stronger increase in places with a higher share of high-income earners (correlation $\rho =$ 0.76), and in places with more owner-occupied rather than rental housing (correlation $\rho = 0.41$ and $\rho =$ -0.42 respectively). Additionally, we show that house values increased stronger in large municipalities ($>$ 100,000 inhabitants) than in smaller municipalities (Table E.4). We will come back to this distinction when describing heterogeneous results by municipalities in Section 5.

4 Empirical Framework

We study how a change in the share of immigrants affects house prices per neighborhood and how this effect differs by income level of immigrants. Our main outcomes of interest are the year-on-year change in the WOZ value and repeat sales index of house prices per neighborhood. We classify immigrants by their annual income corresponding to the quartile of the municipality level income distribution by year.[9] We focus on the year-on-year change in the stock of immigrants relative to the population of a neighborhood, focusing on the demographic composition in the neighborhood.

We argue that the distinction by income provides several channels: firstly, previous research has found that high-income households make neighborhoods more attractive (Duijn and

[9]The categories "low", "midlow", "midhigh" and "high". The category "low" includes non-working immigrants.

Rouwendal (2021)). We want to test whether this distinction holds for immigrants as well. Secondly, in terms of housing demand, individuals with higher income may be more likely to get a mortgage to buy a home, hence exerting upward pressure on house prices through excess demand for housing. Thirdly, the demand for amenities, such as restaurants, may vary with income. This raises the question whether high-income immigrants demand similar amenities also valued by local households. Lastly, immigrant's level of income may influence sorting into neighborhoods and segregation within neighborhoods. While we do not observe household preferences directly, the last two points underline that (native) households may prefer to co-locate with high-income immigrants rather with low-income immigrants. Therefore, we investigate the effect of a change in immigration by income type of immigrants.

For our analysis, we use the following specification:

$$\Delta \log P_{it} = \alpha + \beta \frac{\Delta M_{it}^{j}}{Pop_{it-1}} + \gamma X_{it} + \sigma_i + \delta_t + \varepsilon_{it}. \tag{1}$$

Here, $\Delta \log P_{it}$ is the change in the log measure of prices in postcode i between year t and $t-1$. The main independent variable $\frac{\Delta M_{it}^{j}}{Pop_{it-1}}$ is the change of immigrants in income category j relative to the population in postcode i at time $t-1$. We consider five income categories j, for immigrants: all immigrants regardless of employment status, low, mid-low, mid-high and high income, where the income categories correspond to the quartiles of the municipality level income distribution per year. The coefficient of interest β quantifies the contemporaneous effect of a change in the share of immigrants on the outcome. The variable X_{it} includes time-varying postcode characteristics, such as the change in total population, change in housing stock and unemployment rate to control for housing supply and local economic conditions, respectively. Finally, σ_i and δ_t are postcode and time fixed effects in terms of changes. These control for postcode-specific and national trends affecting house prices, such as national-wide inflation, mortgage conditions or the fact that some postcodes are more popular and house prices appreciate faster across all years. Estimating the equation in first difference eliminates time-invariant factors, such as location and time-invariant neighborhood characteristics as access to public transport. The regression will be weighted by the housing stock to account for size differences across neighborhoods.

A challenge in identifying β is endogeneity of immigrant settlement in neighborhoods. For

example, if immigrants tended to settle in neighborhoods where prices are set to fall, the coefficient β would be negatively biased. To address this problem, we use a shift-share instrumental variable (SSIV) strategy based on historical immigrant patterns. This instrument exploits that immigrants from a country (region) of origin tend to settle in areas where immigrants from the same region have already settled. The instrument then interacts past settlement of immigrants by region of origin with current nation-wide changes in the stock of immigrants by country of origin to predict contemporaneous change in immigrants in a locality. It is widely used in the literature on the economics of immigration as well as in urban and regional economics.[10] We first specify the instrument and then provide descriptive motivation for applying it to the context of this paper.

The instrument is defined as follows:

$$Z_{i,t}^{j} = \sum_{c} \underbrace{\frac{M_{cit_o}}{M_{ct_o}}}_{\text{share}} \quad \underbrace{\frac{\Delta M_{ct}^{j}}{Pop_{it-1}}}_{\text{shift}} \tag{2}$$

where $\frac{M_{cit_o}}{M_{ct_o}}$ denotes the share of immigrants from region of origin c in postcode i at time t_o. The share measures the (past) settlement of immigrants by region of origin across neighborhoods. We use shares fixed in 1995 in the baseline specification and we will argue that results for 10-year lagged shares are virtually the same (Section 5). We use 21 regions of origin as shown in Table E.5.[11] The main reason for clustering is that some countries of origin are hardly represented and a level difference may be over weighted as a percentage (i.e. moving from 5 to 10 immigrants). We also present a robustness check and show that using countries of origin as opposed to regions of origin yields the same conclusion, although with slightly less precision (Section 5).

ΔM_{ct}^{j} is the national-level change in total number of immigrants from region of origin c

[10] The seminal paper is Bartik (1991). Jaeger, Ruist, and Stuhler (2018) provide a list of contributions that use the immigrant settlement shift-share instrumental variable for outcomes such as internal migration (e.g. Card and DiNardo (2000), Wozniak and Murray (2012)), native labor markets (e.g. Card (2001), Ottaviano, Peri, and Wright (2013), Dustmann, Frattini, and Preston (2013), but also female labor market participation (e.g. Farré, González, and Ortega (2011), crime (e.g. Bianchi, Buonanno, and Pinotti (2012), Chalfin (2015)) and firm level adjustments (e.g. Dustmann and Glitz (2015))

[11] We aim to construct groups of origin countries that are similar in size by using the following criteria: visa status (i.e grouping EFTA countries), geographic proximity, language proximity. Some countries are kept as their own category dues to large immigration flows (i.e. Turkey, Indonesia, Germany, Poland).

and income group j. We show the national stock and the year-on-year change of immigrants by income group and country of origin in Figure E.4. Firstly, the stock of immigrants in the Netherlands has been increasing across income groups and countries of origin. Secondly, the stock of low-income immigrants is about twice as big as the stock of high-income immigrants. Thirdly, the year-on-year change in the national-level stock of immigrants, the shift component in the SSIV, varies across country of origin and income group. Changes across income groups are only moderately correlated ($\rho = 0.56$), suggesting that there may be some income-specific driver of the variation in immigration.

The main identification assumption is that the 1995 settlement patterns of immigrants do not affect the change in future house prices. Within the context of the recent advances in the literature of shift-share instruments (for example Borusyak, Hull, and Jaravel (2025)), we consider the shares as exogenous.
Using the above described instrument, the equation for the first stage is:

$$\frac{\Delta M_{it}^{j}}{Pop_{it-1}} = \theta Z_{it}^{j} + \gamma X_{it} + \sigma_i + \delta_t + \nu_{it} \tag{3}$$

We provide further motivation to use the "origin settlement" in our context. This type of clustering is documented across countries and is typically motivated by the existence of immigrant networks that facilitate the process of searching for a job, housing and assimilation to a culture (Munshi (2003)). For the Netherlands, Zorlu and Mulder (2008) show evidence of co-ethnic clustering of immigrant settlement, both for initial and subsequent settlement.

Additionally, we provide two more pieces of descriptive evidence for the distribution of immigrants in our sample of neighborhoods. As a first indicator, we compute the Herfindahl-Hirschmann Index (HHI) to measure the ethnic dominance of each origin group within the local immigrant population. For every region of origin c, we compute the share of immigrants from country c in postcode i relative to all immigrants in that postcode, $\frac{M_{ci}}{M_i}$. The HHI is computed as the sum of squared shares, The HHI is computed as the sum of squared shares, $\text{HHI}_c = \sum_i \left(\frac{M_{ci}}{M_i}\right)^2 \in (0, 1)$. A higher HHI indicates that origin group c constitutes a larger fraction of the immigrant population across many postcodes simultaneously, reflecting stronger ethnic concentration within the local immigrant stock. Table E.6 presents the HHI per region of origin and income group. The results reveal considerable heterogeneity across origin groups.

Small-volume groups such as EFTA and Oceania/Australia/NZ exhibit the highest HHI values (0.69 and 0.56 respectively), indicating they dominate the immigrant composition in the postcodes where they settle. By contrast, large-volume groups such as Poland, Germany, and the Middle East show very low HHI values (0.03–0.05), consistent with broad dispersion across postcodes. Regarding income differences, the evidence is mixed and does not support a general conclusion that low-income immigrants concentrate more strongly than high-income immigrants. While North Africa shows a pronounced difference (0.21 for high-income versus 0.51 for low-income), India displays the opposite pattern (0.14 versus 0.04), and many origin groups show no meaningful difference across income groups.

As a second indicator, we show that the country of origin is a relatively stronger predictor than the income level. We regress the stock of immigrants in year t, neighborhood i and region of origin c on the 10-year lagged stock of immigrants from the same country of origin and the 10-year lagged stock of immigrants with the same income group but any country of origin.

$$N^{j}_{i,t,c} = \lambda_1 N_{i,t-10,c} + \lambda_2 N^{j}_{i,t-10} + \varepsilon_{i,t} \tag{4}$$

with $N^{j}_{i,t,c}$ the number of immigrants with income level (group) j, in neighborhood i, year t and from region of origin c. Table E.7 presents regression results for the current stock of high-income and low-income immigrants on their 10-year lagged stocks, conditioning on postcode and year fixed effects. For both income groups, the 10-year lagged stock of all immigrants from the same country of origin is a strong and significant predictor of current immigrant stocks. For high-income immigrants, the lagged total origin stock explains 43% of current variation, while the lagged high-income stock explains 34%. Notably, the coefficient on the lagged high-income stock (0.135) exceeds that on the lagged total stock (0.078), suggesting that beyond broad co-ethnic networks, high-income immigrants exhibit additional sorting toward postcodes where prior high-income have settled. For low-income immigrants, the lagged total origin stock explains 88% of current variation, versus 28% for the lagged low-income stock, with a substantially larger coefficient on the total origin stock (0.574 versus 0.054), consistent with low-income immigrants following broad co-ethnic settlement patterns strongly. While this cannot strictly be interpreted as causal inference, it serves as motivation to define the shares by country of origin in the instrument (Equation 3). The fraction of explained variation (R^2) indicates that immigrants

are more likely to cluster by region of origin or income group.

5 Results

We first show that the shift-share instrumental variable strategy yields a strong and relevant instrument. After that, we discuss the estimation results for the main outcomes of the property value (WOZ value) and the repeat sales index.

Table 3 shows the estimation results for the first stage, using 1995 shares interacted with national-level year-on-year change of the stock of immigrants by region of origin and income type. Using the shifts per income type, we build an instrument for every income type of immigrants. This instrument is a strong predictor of the contemporaneous year-on-year change of the stock of immigrants per neighborhood, as indicated by a positive and statistically significant coefficients.

Table 3: First stage coefficients

	All migrants (1)	High income (2)	Midhigh income (3)	Midlow income (4)	Low income (5)
$\frac{\Delta M_{it}^{all}}{Pop_{it-1}}$	0.804***	0.905***	0.454***	0.892***	1.305***
	(0.100)	(0.279)	(0.091)	(0.047)	(0.083)
FE: Year	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x
N	24,025	24,025	24,025	24,025	24,025
Adj. R^2	0.587	0.377	0.234	0.314	0.482
IV F statistic	901.0	885.8	287.9	2973.5	1412.1
IV Wald statistic	64.3	10.5	25.1	362.7	248.3

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
First stage, as in Equation (3), with an income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Outcomes measured in 2007 - 2021. Standard errors in parentheses, clustered by municipality. The regression is weighted by the housing stock. "IV F" statistic reports the Cragg-Donald F-statistic under homoskedasticity. "IV Wald" reports the Wald statistic for weak instruments (Kleibergen-Paap rk statistic) accounting for clustered standard errors. Table F.8 shows that the first stage coefficients hardly change when including controls.
Source: *CBS Microdata*

The R^2 reflects a substantial fraction of explained variation. It is between 0.23 and 0.58, depending on the type of immigrant. Additionally, the F-statistic in the first stage is very high,

above 200. We also present the IV Wald F-statistics (Kleibergen-Paap statistic, Kleibergen and Paap (2006)) which account for clustered standard errors and test for weak instruments. While the latter F-statistics are smaller than the regular F-statistic, they are above the typical threshold of 10, underlining a strong and significant instrument. In line with higher clustering of low-income immigrants previously discussed, we note that the instrument is a stronger predictor for low-income immigrants.

Table 4 and Table 5 report second-stage IV estimates of the effect of immigration on property values, measured by the WOZ administrative valuation, and on the Case-Shiller repeat sales index respectively. The identifying variation comes from the shift-share instrument, which isolates changes in immigrant inflows driven by the historical settlement patterns of national-origin groups interacted with aggregate sending-country flows. All specifications include year and 4-digit postcode fixed effects; even-numbered columns additionally control for time-varying local unemployment, housing stock changes, and population changes.

The central finding of this paper is that the effect of immigration on property values is heterogeneous by the income profile of the arriving immigrants, and that this heterogeneity is not visible in the pooled estimate. A one percentage point increase in the overall immigrant share (column 1) is associated with a modest and imprecisely estimated increase in property values across both outcomes. Decomposing by income group, however, reveals a clear and economically meaningful gradient: inflows of middle-income immigrants — particularly those in the middle-low income bracket — are associated with significant property value appreciation, while inflows of low-income immigrants are associated with a statistically significant decline in property values. This sign reversal might suggest that the aggregate null result masks compositional heterogeneity: immigration raises property values when it brings workers who raise local labor market quality and consumption demand, but reduces them when the inflow is concentrated in the lowest income tier, consistent with neighborhood filtering dynamics and a deterioration in the amenity bundle that existing and prospective residents value.

We now turn to the individual income group estimates, reported in columns (3)–(10) of Table 4. A one percentage point increase in the high-income immigrant share is associated with a 8.3 percentage point increase in WOZ value, but the estimate is imprecise — significant only at the 10% level under 3-digit postcode clustering and nonsignificant under municipality clustering. The wide confidence interval means we cannot rule out effects ranging from negligible

Table 4: Second stage coefficients for property value (WOZ)

	Dependent variable: Lead of WOZ value									
	All		High income		Midhigh income		Midlow income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\frac{\widehat{\Delta M_{it}^{j}}}{Pop_{it-1}}$	1.188	2.165	8.309	9.362	15.384	16.054	7.484	8.584	-1.810	-4.881
	(1.973)	(2.760)	(10.790)	(11.189)	(9.661)	(10.594)	(0.444)***	(0.955)***	(2.078)	(3.338)
	[0.523]***	[0.687]***	[5.415]	[5.626]*	[6.147]**	[6.336]**	[0.594]***	[0.743]***	[1.017]*	[1.587]***
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776
Adj. R^2	0.648	0.637	0.624	0.621	0.382	0.379	0.611	0.602	0.634	0.538
Avg outcome	1.9%	1.9%	1.9%	1.9%	1.9%	1.9%	1.9%	1.9%	1.9%	1.9%
Avg regressor	0.19%	0.19 %	0.038 %	0.038 %	0.029 %	0.029 %	0.036 %	0.0364 %	0.086 %	0.086 %
SD regressor	0.00268	0.00268	0.000534	0.000534	0.000657	0.000657	0.001014	0.001014	0.001526	0.001526
Impact at mean regressor	0.23%	0.42%	0.32%	0.36%	0.44%	0.46%	0.27%	0.31%	-0.14%	-0.39%

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Controls include the year-on-year change on the unemployment rate, housing stock and total population. The outcome is measured in 2007 - 2021. Standard errors clustered by municipality are in round brackets, standard errors clustered by 3-digit postcode are in square brackets.
Source: *CBS Microdata*

to economically large, and we interpret this estimate cautiously. One plausible mechanism is that high-income immigrant inflows raise local consumption of tradeable amenities — restaurants, retail, professional services — in ways that capitalize into property values, but that this channel operates slowly or unevenly across neighborhoods, contributing to imprecision.

Middle-high income immigrants generate a more precisely estimated positive effect. A one percentage point increase in their share raises annual WOZ growth by approximately 15.4 percentage points, significant at the 5% level under both clustering schemes and stable across specifications. This estimate corresponds to roughly 0.44 percentage points at the mean inflow — approximately one-fifth of mean annual appreciation. A plausible mechanism is labor market complementarity: middle-high income immigrants may fill occupational niches that raise local productivity and, in turn, housing demand among incumbent residents.

The strongest and most precisely estimated positive effects arise from middle-low income immigrants. A one percentage point increase in their share is associated with a 7.5–8.6 percentage point increase in WOZ value (columns 7–8), significant at the 1% level under both clustering schemes. At the mean inflow this corresponds to 0.27–0.31 percentage points — again, roughly

one-sixth of mean annual appreciation. Economically, middle-low income immigrants likely increase local labor supply in service and construction sectors, sustaining housing demand and neighborhood commercial activity without triggering the amenity deterioration associated with very low-income concentration.

Low-income immigrant inflows stand in contrast to the other findings. A one percentage point increase in low-income immigrant share is associated with a 1.8-4.9 percentage point decline in WOZ value (columns 9–10), translating to a reduction of 0.14–0.39 percentage points at the mean. The estimate is significant at the 10% level in the baseline and at the 1% level once controls are added, and its magnitude grows with the inclusion of controls — suggesting that if anything, omitted local demand factors attenuate the negative effect. This pattern is consistent with a neighborhood filtering mechanism: concentrated low-income immigrant inflows may signal or accelerate neighborhood transition, reducing the willingness to pay of existing and prospective higher-income residents. It may also reflect amenity deterioration, changes in school composition, or increased perceived crime risk.

Columns (2), (4), (6), (8), and (10) augment the baseline with time-varying local controls. Across all income groups, point estimates are stable or marginally larger in magnitude — a pattern consistent with the instrument being orthogonal to local demand and supply conditions. The stability is particularly notable for the middle-low and low-income estimates, where concerns about omitted time-varying factors are greatest given the geographic concentration of these immigrant groups.

We report two clustering schemes to account for spatial correlation in the residuals. Standard errors clustered at the municipality level appear in round brackets; those clustered at the 3-digit postcode level appear in square brackets.[12] Municipality-level standard errors are consistently larger, reflecting within-municipality correlation in residuals that postcode clustering does not fully absorb. Our preferred baseline is municipality clustering, which is the more conservative choice. All qualitative conclusions hold under both schemes, with the exception of the high-income estimate in column (3), which loses significance under municipality clustering. We flag this explicitly and treat the high-income result as suggestive rather than conclusive.

Table 5 presents estimates using the Case-Shiller repeat sales index as the outcome. We treat

[12]In our sample, a 3-digit postcode contains on average 3.3 four-digit postcodes (maximum: 9); a municipality contains on average 6 (maximum: 60, for Amsterdam).

these as a validation exercise rather than a primary result. The repeat sales index is constructed from repeated property-level transactions, making it a market-based, transaction-price measure of house price growth that is free from the administrative smoothing inherent in the WOZ valuation. In principle, this makes it a sharper signal of true market conditions. In practice, however, it is subject to two important limitations in our setting. First, because it requires at least two transactions on the same property, it is based on a selected subsample of properties, and the selection may be correlated with neighborhood change — precisely the margins we are studying. Second, as shown in Table 2 and Appendix Figure E.1, the index exhibits substantially greater cross-neighborhood dispersion and lower average year-on-year variation than the WOZ measure, generating less statistical power.

Table 5: Second stage coefficients for repeat sales index

	Dependent variable: repeat sales index									
	All		High income		Midhigh income		Midlow income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\frac{\widehat{\Delta M_{it}^j}}{Pop_{it-1}}$	0.474	-0.245	1.540	-0.043	3.006	2.992	3.341	4.456	-1.326	-5.697
	(1.018)	(0.570)	(4.339)	(2.622)	(2.678)	(1.967)	(0.701)***	(0.794)***	(1.177)	(1.291)***
	[0.840]	[0.744]	[2.986]	[2.167]	[3.659]	[3.568]	[0.881]***	[0.907]***	[0.750]*	[0.979]***
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776
Adj. R^2	0.124	0.131	0.124	0.130	0.120	0.126	0.121	0.124	0.115	0.057
Avg outcome	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Avg regressor	0.19	0.19	0.038	0.038	0.029	0.029	0.036	0.0364	0.086	0.086
SD regressor	0.00268	0.00268	0.000534	0.000534	0.000657	0.000657	0.001014	0.001014	0.001526	0.001526
Impact at mean regressor	0.09	-0.05	0.06	-0.001	0.09	0.09	0.01	0.01	-0.11	-0.48

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Controls include the year-on-year change on the unemployment rate, housing stock and total population. The outcome is measured in 2007 - 2021. Standard errors in parentheses, clustered by municipality.
Source: *CBS Microdata*

With these caveats in mind, the repeat sales results validate the qualitative pattern from the WOZ analysis. Middle-low income immigrant inflows again generate the strongest and most precisely estimated positive effect, significant at the 1% level under both clustering schemes, while low-income inflows are associated with declining house price growth, significant at the 10%–1% level depending on specification. Two notable departures from the WOZ results deserve

attention. The pooled estimate (column 2) and the high-income estimate (column 4) change sign upon inclusion of controls, becoming negative. This suggests that year-on-year variation in the repeat sales index carries a macroeconomic component that is correlated with aggregate immigration flows but not with income-specific flows — consistent with the index's greater sensitivity to business cycle conditions. We do not view these sign reversals as evidence against our main findings; rather, they reflect a known limitation of the repeat sales outcome in settings with thin or cyclically sensitive transaction volumes. The income-group-specific estimates, where statistical power is concentrated and where the instrument variation is most relevant, tell a coherent and consistent story across both outcomes.

5.1 Robustness to choice of shift and shares

In the following, we show that the estimated coefficients are robust to the last step of our sample selection (Section 3, Table 1). Moreover, we provide results for two modifications of the shift-share instrumental variable. First, we alter the definition of shares and use 10-year lag instead of 1995 as base year. Second, we modify the shift component and use all countries of origin instead of aggregated regions of origin for the shifts.

Table 6 shows the coefficients of the second stage with standard errors clustered by municipality. Column (2) captures the results for a small modification of the sample selection as outlined in Section 3 and Table 1: using the 20th percentile in 1995 as a cut-off instead of the 30th percentile. This effectively includes a slightly larger number of (smaller) postcodes to the sample. The estimated coefficients are virtually unchanged compared to the baseline sample in column (1), corroborated by the regressions being weighted by the stock of housing. This underlines that we do not induce major bias when selecting our baseline sample. Columns (3) and (4) show modifications of the SSIV. Firstly, we calculate the 10-year lag shares, and measure the outcome in 2007 - 2021 as in the baseline results. The estimated coefficients for the 10-year lagged shares are very similar to the baseline coefficients. We argue that a 10-year lag is likely to be "far enough" lagged such that the shares are plausibly exogenous to (future) changes in house prices. Secondly, we alter the classification of regions of origin and use countries of origin instead of custom grouping. Overall, estimated coefficients are slightly smaller in magnitude, but the same pattern and direction persist.

Table 6: Robustness of SSIV

$\frac{\widehat{\Delta M_{it}^{j}}}{Pop_{it-1}}$	*Dependent variable: Lead of WOZ value*			
	Baseline (1)	Sample 20th Pct (2)	1995 shares (3)	Country-level shifts (4)
j = all	1.188	1.190	1.188	-0.071
	(1.973)	(1.960)	(1.784)	(2.346)
j = high income	8.309	8.358	7.375	3.333
	(10.790)	(10.785)	(8.880)	(6.396)
j = midhigh income	15.384	15.221	15.487	8.083
	(9.661)	(9.469)	(9.737)	(8.311)
j = midlow income	7.484***	7.466***	6.628***	6.645***
	(0.444)	(0.446)	(0.521)	(0.349)
j = low income	-1.810	-1.796	-1.295	-1.676
	(2.078)	(2.070)	(1.814)	(2.788)
FE: Year	x	x	x	x
FE: 4-digit postcode	x	x	x	x
N	24,555	25,901	24,555	24,555

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable. One column per sample, one row per regression. The outcome is measured in 2007 - 2022. Standard errors in parentheses, clustered by municipality. Column (1), (3) and (4) are estimated on the same sample of postcodes, but columns (3) and (4) present a variation in the SSIV.
Source: *CBS Microdata*

5.2 Heterogeneity by areas

Next, we turn to heterogeneity by sub-samples of geographic areas. The baseline sample contains neighborhoods across the entire country. However, housing and immigration dynamics may differ by region, for example in cities or less densely populated areas. We split the sample in two sets: Firstly, postcodes in a municipality with at least 100,000 inhabitants in 2021, reflecting more urban and more densely populated areas. This distinction is supposed to capture a broad set of relatively urban and densely populated postcodes (N = 482) in 18 municipalities. In the following, we refer to municipalities with > 100,000 inhabitants as "urban" and less urban vice versa. Secondly, we distinguish the biggest four cities (Amsterdam, Rotterdam, The Hague and

Utrecht). This distinction follows Hekwolter, Nijskens, and Heeringa (2017) assessment that housing markets in the cities experience different dynamics than the rest of the country.

We present descriptive statistics for urban and less urban municipalities (by less or more than 100,000 inhabitants) in Appendix Table E.4 and highlight the main distinctions here. On average, postcodes in less urban municipalities contain a smaller population and fewer immigrants, resulting in a substantially smaller share of immigrants as well. Postcodes in less urban municipalities contain on average 5,301 inhabitants and 4,055 dwellings per 4-digit postcode compared to an average of 7,272 inhabitants and 6,226 dwellings per postcode in urban municipalities. The immigrant share is 27% in "urban" municipalities, while it is on average 13% in postcodes in less urban municipalities. While the housing stock is smaller in less densely populated postcodes, average household size is larger. Lastly, the share of owner-occupied units is around 50% higher in urban municipalities at 62% on average versus 43% in urban municipalities.

We show the results for this heterogeneity exercise for our main outcome WOZ value in Table 7. We find that the effects are mainly driven by postcodes in less densely populated areas (municipality with $<$ 100,000 inhabitants and not in "Big 4 city"). Second stage estimates for the year-on-year change of the WOZ value are small and statistically insignificant for all types of immigrants in the set of postcodes in the four big cities and urban municipalities. For the set of postcodes outside the four big cities, the estimated coefficients are larger in size and statistically significant.

Potential explanations for the heterogeneity of the effect of immigrant changes on property values include the structure of the housing market and demographic composition. Firstly, the housing market in the large cities may be inefficient or distorted for reasons beyond immigration (assessed as "overheating" as in Hekwolter, Nijskens, and Heeringa (2017)). This is reflected by fast transactions above asking price, indicating that housing supply is tight. With this dynamic, the main drivers of the housing market in large cities may be beyond immigration or general demographic change. It is notable from the descriptive statistics that the average year-on-year change in the WOZ values is 3.6% in postcodes in urban municipalities while it is 2% in less urban municipalities. Second, attitudes towards immigration may vary by location. Relatedly, immigration is less common in the less urban areas and thus a change is more salient. This is reflected in a lower immigrant share in population in the descriptive statistics (Table E.4). We show some indications from the World Values Survey and how they vary by the size of

Table 7: Second stage coefficients for different samples

	Dependent variable: Lead of WOZ value				
$\frac{\widehat{\Delta M_{it}^{j}}}{Pop_{it-1}}$		Municipalities		Cities	
	Baseline (1)	>100k (2)	< 100k (3)	Big 4 (4)	< Big 4 (5)
j = all	1.188	0.012	4.435***	0.799	2.654***
	(1.973)	(2.433)	(1.044)	(3.053)	(0.945)
j = high income	8.309	1.369	20.992***	-3.731	22.736***
	(10.790)	(12.222)	(6.393)	(12.522)	(4.138)
j = midhigh income	15.384	9.998	13.001***	1.896	20.049**
	(9.661)	(17.651)	(3.162)	(27.943)	(9.059)
j = midlow income	7.484***	5.222***	5.851***	2.563	5.883***
	(0.444)	(1.030)	(1.499)	(2.467)	(0.675)
j = low income	-1.810	-1.521	2.002**	-0.154	0.180
	(2.078)	(2.235)	(0.874)	(1.706)	(0.766)
FE: Year	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x
N	24,555	6,489	18,066	1,979	22,576
N Postcodes	1794	482	1312	153	1641
N Municipalities	296	18	278	4	292

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Second stage coefficients, as in Equation (1), with an income-type specific SSIV variable, using 1995 shares. One column per sample. The outcome is measured in 2007 - 2022. Standard errors in parentheses, clustered by municipality.
Source: *CBS Microdata*

town in Table G.10. Due to the small sample size and that the responses are not linked to our sample, these are to be interpreted as an indication. When asked which groups survey participants would *rather not* have as neighbors, 19.8% of (187) respondents in a town size of 10,000 - 20,000 mention immigrants or foreign workers. This percentage is smaller (10.1%) for respondents in large cities of > 500,000 inhabitants. This gradient of less openness to different types of neighbors in small cities is similar for the variables "people of a different race", "people who speak a difference language" and "people of a different religion". The larger the town, the

less survey participants mention that they would rather not have different groups as neighbors. Vice versa, respondents in larger cities more often report to trust people of another nationality and less often agree that natives should be given priority of employment over immigrants. While being a small survey and not directly matched to the neighborhoods in the remainder of this paper, the survey responses from the World Values Survey indicate that there may be a correlation between the place of residence and openness to immigration. A similar variation between location, individual characteristics and attitudes towards immigration is described in Card, Dustmann, and Preston (2005) using the European Social Survey.

Thirdly, the structure of the housing market may affect the speed of adjustment to a change in the demographic composition of a neighborhood. For example, if an increase in the share of immigrants triggers an outflow of natives, inhabitants of owner-occupied units need to sell and purchase a new unit to move away, as opposed to renters that can move without the unit actually being sold. Relocation of homeowners may take some time to realize, which may delay the effect. Whether and which inhabitants move away upon an inflow of immigrants will be subject to future research.

5.3 Results for the medium term

To examine whether the contemporaneous effects identified in Table 4 persist or reverse over time, Table 8 replaces the dependent variable with the 3-year lead of the WOZ change, while keeping the regressor and instrument unchanged — the year-on-year change in immigrant share instrumented by the shift-share instrument. This specification asks a precise dynamic question: does a one-year immigrant inflow shock in year t continue to affect property values through $t+3$, or does the initial price response unwind? The answer is unambiguous: across all income groups, the estimated 3-year lead effects are negative, reversing the positive contemporaneous estimates in Table 4.

The pooled estimate implies that a year-on-year inflow shock reduces property value growth over the subsequent three years by 4.4 percentage points in the baseline and 10.5 percentage points once controls are included, both significant at the 1% level. Evaluated at the mean inflow, these translate to reductions of 1.1 and 2.6 percentage points respectively — economically meaningful relative to the mean 3-year WOZ appreciation of 6.3%. The income-group

Table 8: Effect of a change of immigrant share on 3-year difference in WOZ value

	Dependent variable: 3-year change WOZ value									
	All		High income		Midhigh income		Midlow income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\frac{\widehat{\Delta M_{it}^{j}}}{Pop_{it-1}}$	-4.370***	-10.459***	-26.862	-31.757	-37.294	-37.387	-5.869	-5.653	-4.623***	-20.284***
	(6.487)	(8.371)	(23.109)	(22.714)	(43.768)	(42.982)	(14.099)	(16.139)	(1.591)	(5.107)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	16,105	14,845	16,105	14,845	16,105	14,845	16,105	14,845	16,105	14,845
Adj. R^2	0.844	0.812	0.822	0.815	0.679	0.713	0.862	0.866	0.848	0.635
Avg outcome	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%
Median outcome	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%	7.8%
Avg regressor	0.245%	0.245%	0.044%	0.044%	0.054%	0.054%	0.054%	0.054%	0.049%	0.049%
SD regressor	0.00284	0.00284	0.00055	0.00055	0.00071	0.00071	0.00071	0.00071	0.001	0.001
Impact at mean regressor	-1.07%	-2.56%	-1.18%	-1.39%	-2.01%	-2.02%	-0.32%	-0.32%	-0.22%	-0.99%

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Outcome: year-on-year change in the 3-year change of WOZ value per neighborhood (4-digit postcode). Second stage coefficients, as in Equation (1), with an income-type specific SSIV variable, using 1995 shares. The outcome is measured in 2007 - 2022. Standard errors in parentheses, clustered by municipality.
Source: *CBS Microdata*

decomposition yields uniformly negative point estimates, though only the low-income estimates are precisely estimated. The baseline coefficient of −4.6 implies that, at the mean annual low-income inflow of 0.049 percentage points, property value growth over the subsequent three years is reduced by 0.22 percentage points — approximately 3.5% of mean 3-year appreciation. The controlled estimate of −20.3 (column 10) implies a substantially larger reduction of 0.99 percentage points, or 15.7% of mean appreciation, suggesting that omitting local demand controls attenuates the negative dynamic effect considerably. The combination of positive contemporaneous and negative 3-year lead estimates is consistent with an incumbent displacement mechanism. An immigrant inflow raises housing demand on impact, generating the positive year-on-year price response documented in Table 4. Over the subsequent three years, however, the inflow appears to trigger outmigration among existing higher-income or longer-established residents — a neighborhood transition dynamic that erodes the local demand base and reverses the initial price appreciation. This interpretation aligns with a well-established literature on demographic neighborhood change and housing markets (Boustan (2010) and Saiz and Wachter

(2011)), and generates a testable prediction: incumbent outmigration should rise in the three years following an inflow shock. We also note that the reversal is sharpest for the pooled and low-income estimates — precisely the groups where neighborhood composition effects are most plausible — while the mid-income positive effects documented year-on-year are less precisely reversed at the 3-year horizon, suggesting some heterogeneity in the speed and completeness of the adjustment process. One methodological note: the 3-year lead specification requires that the shift-share instrument in year t is uncorrelated with confounders affecting property values in $t + 3$ through channels other than the immigrant inflow. Given that the instrument is constructed from national-origin settlement patterns that predate our sample by a decade, and that we condition on postcode and year fixed effects, this assumption is plausible. Nevertheless, we interpret the dynamic estimates with appropriate caution and treat them as motivating evidence for the displacement mechanism rather than a definitive causal estimate of the long-run effect.

5.4 Extensions: housing stock and population changes

In the following, we turn to two important factors of the housing market, namely the total change in population and the housing stock. We briefly describe how neither housing stock nor total population changes markedly in response to immigration. This supports that our estimated effects are due to demographic composition in the neighborhood as opposed to shifts in demand and supply the housing market.

Firstly, we turn to the supply side of the housing market. We see from the descriptive statistics in Table 2 that the year-on-year change in the housing stock is essentially zero. Note that we only include neighborhoods that have existed for the entire duration 1995 - 2021 and have a non-zero number of inhabitants. This implies that any change in the housing stock occurs within existing neighborhoods in our sample and we do not consider newly built neighborhoods or development projects.[13] To some extent, this stands in contrast to findings from Spain (Gonzalez and Ortega, 2013), who find that an inflow of predominantly low-income immigrants is associated with a 1.2-1.5% increase in housing units through construction activity and a labor supply shock (of immigration) in the construction sector. We argue that the "construction channel" is less important in the context of this study. We consider the entire income distribution

[13] We describe how we select the sample of neighborhoods in Section 3 and Table 1.

of immigrants and therefore include a large range of occupations and income levels in our sample. This contrasts to a large inflow of low-educated (low-income) immigrants in Spain as considered by Gonzalez and Ortega (2013). Beyond labor supply, potential constraints to the construction sector also include land, capital or building permits. For the context of this study, the main point is that the supply side of the housing market is virtually unchanged in our sample.

Secondly, we turn to the demand side of the housing market. We run the same regression as before (Equation 1), with the outcome being the year-on-year change of population in a neighborhood. The regression reflects whether a change in the immigrant share also increases total population in a neighborhood. A net increase in population per neighborhood would then contribute to a mechanical increase in house prices through increased demand for housing. Table 9 displays the estimated effect for a change in the immigrant share on the net change of year-on-year population, including both local (native population) and immigrants. The estimated coefficients are quite small (< 1) for all types of immigrants and mostly statistically insignificant (Column (1), (3), (5), (7) and (9)). Including controls for the stock of housing and unemployment further reduces the magnitude of the coefficients (uneven numbered columns). We conclude that the total population per neighborhood is likely unchanged upon a change in the share of immigrants. Given that the total population is unchanged, but the immigrant share increases, there must be some native (out-) mobility as well.

6 Discussion and conclusion

In this paper we investigate how a change in the immigrant share per neighborhood affect property valuation (house prices) in the Netherlands. We distinguish immigrants by their income, relative to the municipality-level income distribution. This further nuances findings from preceding literature, which mostly treats immigrants as a homogeneous group.

We use comprehensive administrative data for the entire population and residential addresses in the Netherlands. We conduct our analysis on the neighborhood level (4-digit postcode), which contains around 4,300 properties and a population of around 5,500 inhabitants. To circumvent the endogeneity of immigrant location choice, we use a origin-settlement shift-share instrumental variable strategy. This instrument uses past shares of immigrants per neighborhood and interacts it with current year national-level immigrant inflows. We show descriptively that the instrument

Table 9: Effect of a change of immigrant share on a change in total population

	Dependent variable: Total population									
	All		High income		Midhigh income		Midlow income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\frac{\widehat{\Delta M_{it}^{j}}}{Pop_{it-1}}$	0.333	0.008	0.679	0.217	-0.301	0.163	-0.302	0.062	0.636***	-0.064
	(0.360)	(0.041)	(1.385)	(0.184)	(1.010)	(0.341)	(0.198)	(0.182)	(0.213)	(0.123)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	23,335	22,979	23,335	22,979	23,335	22,979	23,335	22,979	23,335	22,979
Adj. R^2	0.341	0.845	0.209	0.845	0.166	0.845	0.168	0.845	0.395	0.841

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Outcome: year-on-year change in total population per neighborhood (4-digit postcode). Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. The outcome is measured in 2007 - 2022. Standard errors in parentheses, clustered by municipality.
Source: *CBS Microdata*

is strongly predictive of the change in the immigrant share per neighborhood. Additionally, we show that the results are robust to modification of the instrument.

Our results reflect three main conclusions. Firstly, the contemporaneous effect a change in the immigrant share on property values varies with immigrant income. For the entire set of immigrants, we find a positive effect of 0.23% point on year-on-year change of property value. This effect is positive for immigrants with income in the 3 upper quartiles of the income distribution, but turns negative for low-income immigrants (lowest quartile). Secondly, we show that these effects are mainly driven by neighborhoods in less urban municipalities ($<$ 100,000 inhabitants). This may be rooted in differences in the housing market, demographic structure and a lower baseline immigrant share. Additionally, we show that survey respondents in the *World Value Survey* are less open towards immigrants in less urban areas. Thirdly, we also consider medium-term results using a 3-year change in the outcome. For this time frame, our results turn negative for all types of immigrants. Relative to a 6.3% average 3-year change in property values, the average effect of a change in the immigrant share is -1.07% percentage points. We argue that this negative result may be driven by (lagged) native out-mobility.

The results are largely in line with the findings from the literature. In the short run (year-

on-year) and in narrowly defined areas, most contributions find a positive effect of immigration on house prices, arguably demand-driven (Mussa, Nwaogu, and Pozo (2017), Accetturo et al. (2014), Saiz (2003)). Some studies consider native out-mobility and a longer time-horizon and tend to find negative effects mediated by native out-mobility from areas receiving immigration (Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019), Sá (2015)).

Which households move out and where they go will be a future topic for research. A first contribution in this direction is Boje-Kovacs et al. (2024) who consider detailed native responses on the neighborhood and even building level to an inflow of non-western refugees. A distinction for immigrant income and characteristics is worthwhile to consider in this direction. Some studies in urban economics suggest that households prefer to reside in neighborhoods with (other) high income households (e.g. Duijn and Rouwendal (2021)). Whether this preference also holds for (high-income) immigrants is still open. Another area for future research is the distinction between rental prices and house values (transaction prices). In this paper, we consider property valuation and transaction prices in the absence of population wide information on rents. Depending on turnover of renters, rental prices may be more flexible to adjust in the short run and are another margin

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A Appendix

Table A.1: Literature overview, most to least recent

Authors	Country	Region	Type of migrant	Main result
Boje-Kovacs et al. (2024)	Denmark, 1987-2017	neighborhoods building unit	non-Western, refugees	0.3% ↓ house prices
Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019)	Spain 1998-2008	neighborhood (census tracts, N = 28,523)	foreign-born, excl. children	0.33% ↑ native outflow; 50% ↑ construction
Mussa, Nwaogu, and Pozo (2017)	US, 2002 - 2012	MSA (N = 275)	foreign-born	0.8% ↑ house prices 0.8% ↑ rental prices 9.6% ↑ house prices in surrounding MSAs 1.6% ↑ rent in surrounding MSAs
Sá (2015)	UK, 2003-2010	local authorities (N =170)	foreign-born, 16-65 years	1.7% ↓ house prices; 0.05% ↑ native net out-migration
Gonzalez and Ortega (2013)	Spain, 2000-2010	provinces (N = 50)	foreign-born, working age	2% ↑ house prices; 1.2-1.5% ↑ construction of new houses
Accetturo et al. (2014)	Italy, 2003-2010	20 large cities & districts	foreign-born	0.5% ↑ house prices; 0.2% ↓ house prices within district; 0.6 ↑ outflow of natives
Akbari and Aydede (2012)	Canada 1996, 2001, 2006	Census Metropolitan Area (N = 258)	immigrants, recent & settled	0.1-0.12% ↑ house prices; from migrants arrived > 10 yrs ago
Saiz and Wachter (2011)	US, 1980, 1990, 2000	MSA (N = 67) Census tracks	foreign-born	0.134% ↓ native population; 0.678% ↓ non-Hispanic-white population 6% ↓ in housing values
Saiz (2003)	US, 1974 - 1983	MSA, Miami and Florida	Mariel boatlift	8-11% ↑ in units of low quality; 8% ↑ additional for units inhabited by low income Hispanic residents in 1979 than by non-Hispanic

B Data appendix

We combine datasets on individuals, house transactions and values and neighborhoods from Statistics Netherlands (CBS) and Land Registry (Kadaster). These datasets cover the whole territory of the Netherlands. Demographic information is available for every resident in the Netherlands: date and place of birth, country of origin, gender, parent's dates and places of birth and country of origin. Immigrants living in the Netherlands for at least four months are required to register the address of residency at the Dutch municipality, together with the date of arrival and departure and the type of migration.[14] The linked employer-employee dataset contains information on their monthly wage, gross and net, hours worked and dates of employment for regularly employed individuals. Due to the registration requirement, every individual is linked to their house address and dates of start and end of residence. If the house has been sold, for every sale the Land registry records transaction price, date, house value at the time of the transaction, recent house value, tenure type.
Characteristics of neighborhoods are reported in *Kerncijfers wijken en buurten* (KWB) for the year 1995, 1997, 1999, 2001, and from 2003 onwards. In these datasets, the characteristics are reported at municipality, district (*wijk*) and neighborhood (*buurt*) levels. We aggregate the data of neighborhoods (*buurt*) to 4-digit postal code level. A postcode might contain more than one neighborhood.

C Postcode selection

In the following, we describe in detail the selection of postcodes. Using comprehensive data on population per postcode from 1995 - 2022, the goal is to construct a balanced panel of postcodes. The variables of interest for this study are the (change in) the stock of migrants and the stock of population.

Firstly, 4061 4-digit postcodes are consecutively available between 1995-2022. In the next steps, we exclude postcodes that ever have a registered population of 0 inhabitants or migrants,

[14]There are four migration types: 1) Immigration, settlement from abroad; 2) Administrative inclusion, resettlement following a removal of the person without them ever leaving the Netherlands; 3) Emigration, removal of a person from the registry when departing abroad for at least 8 months; 4) Administrative discharge, removal from registry when the person is unreachable. We select immigrants type 1).

Table C.2: Moments of population, migrant population and stock of houses by sample restriction

Filtering Step	N postcodes	Inhabitants			Migrants			Housing Stock		
		25th Pct.	Mean	75th Pct.	25th Pct.	Mean	75th Pct.	25th Pct.	Mean	75th Pct.
All postcodes	4061	486	3291	5446	486	3291	5446	486	3291	5446
> 0 population	2980	784	3912	6299	784	3912	6299	784	3912	6299
> 0 migrants	2802	923	4035	6417	923	4035	6417	923	4035	6417
Growth rate inhabitants	2751	941	4065	6455	941	4065	6455	941	4065	6455
Growth rate migrants	2671	1045	4169	6547	1045	4169	6547	1045	4169	6547
10th pct population (1995)	2360	1609	4642	6915	1609	4642	6915	1609	4642	6915
20th pct population (1995)	2075	2479	5197	7342	2479	5197	7342	2479	5197	7342
30th pct population (1995)	1795	3409	5801	7746	3409	5801	7746	3409	5801	7746

Note: Each line represents a successive sample selection. Population and number of migrants both refer to individuals age ≥ 18. Housing stock is defined as the number of dwellings with a registered address. Postcode refers to a 4-digit postcode.
Source: *CBS Microdata*

with the reference point being 1 January. Next, we turn to a restriction on year-on-year changes in the stock of inhabitants and migrants. We exclude postcodes that exhibit an absolute change of $> 100\%$ in those variables, as they may be outliers in the analysis. The number of postcodes excluded from this step is small, around 50-70 postcodes. The last restriction on the sample is the stock of inhabitants in 1995, the first year of observations. We exclude very sparsely population postcodes, as we are interested in changes in the demographic composition. Therefore, we present a sample selected on a minimum threshold of inhabitants, defined as the 10th, 20th and 30th percentile of the population distribution in 1995.

D Repeat sales index

The repeat sales index represents the price appreciation in a given area over time. Using repeated sales of the same property allow to control for property-specific time-invariant characteristics. We obtain the repeat sales index following Bailey, Muth, and Nourse (1963). The model is given by:

$$r_{itt'} = \sum_{j=1}^{T} \beta_j x_j + u_{itt'} \tag{5}$$

where $r_{itt'}$ is the natural logarithm of the ratio of the final sales price in year t' and the initial sales price in year t for each transaction pair of dwelling i. The variable x_j takes value -1 if time

j is the year of initial sale, 1 if j is the year of final sale, and 0 in all other periods. The OLS estimate $\hat{\beta}_j$ gives the log price aggregate index in year j.
To estimate the regression, we use the set of properties sold at least twice from 1995 to 2023. This yields a number of transaction pairs for house i equal to the number of times the house has been sold minus 1.
First, we create dummy variables x_j, then the natural logarithm of the ratio of final and initial sales prices. Finally, we estimate the regression for each dwelling i. The coefficient $\hat{\beta}_j$ represents the house price index per 4-digit postcode.
Almost half of the dwellings sold in our sample have been transacted only once. Among the 4-digit postcodes in the 30 selected municipalities, there are 48 postcodes with houses sold only once between 1995 and 2023. Within postcodes there is a lot of heterogeneity in how many years pass between two transactions for the same house. The range goes from 0 to 34. On average across postcodes, the average sale time of a house is 6 years, with the 75th percentile being 8 years. Thus, houses sold once every 25 years are outliers.
The drawback of this methodology is that the variance increases with the interval of time between transactions. Hence, the OLS estimate is not efficient (Ghysels et al., 2013). The houses sold more than twice may not be representative of the whole sample, leading to a selection bias.
A possible solution for this issue is offered in Case and Shiller, 1987. In their specification, they include a term to capture the imperfections of the housing market. Also, they have a component increasing with the length of the holding interval. Their estimation approach has three steps:

1. Estimate regression (5) and store residuals

2. Weighted list square regression of the residuals squared on a constant and on the time interval between transactions

3. GLS regression of (5), where the weights are the inverse of the square root of the fitted values from point 2.

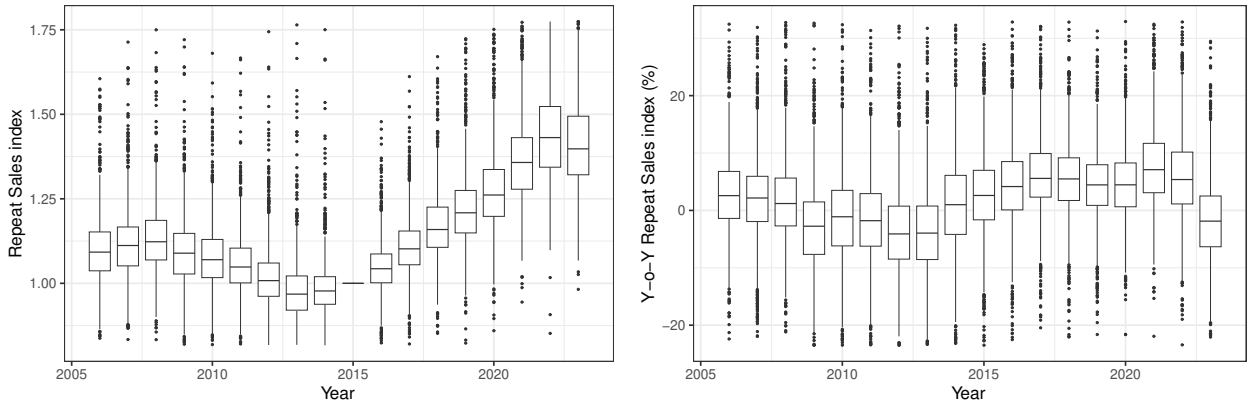
Moreover, the repeat sales index does not consider what might determine the price of a property at a given point in time.

Then, we calculate the repeat sale index over time per postcode, using the Case and Shiller method. The steps are the following:

- Select a postcode and keep only the unique combinations of property identifier and sale date.
- Keep properties that are sold more than once If the subset for that postcode has more than 10 observations – this could imply that there could be few houses sold multiple times – proceed as follows for every year and postcode:
- Create a dummy variable per year from 1995-2023 and per property unit identifier
- Per unit, if the earliest sale date is equal to the dummy year, assign -1, this is because the first sale date should have -1. If the latest date is the same as the dummy year, assign 1. Otherwise 0.
- For houses with more than 2 sales date, per property identifier, calculate the following ratio: $log(\text{sale price}_t/\text{sale price}_{t-1})$
- Run the regression: $log(\text{sale price}_t/\text{sale price}_{t-1}) \sim$ dummy
  - Calculate the interval between sales year
  - Take the square of the residuals
  - Run the regression: squared residuals on time interval with constant
  - Store the fitted values
- For the third step: store the inverse of the square root of the fitted values: $\frac{1}{\sqrt{(fittedvalues)}}$
- Run the regression: ratio on all dummies per year with weights the inverse residuals, without constant.
- Save the exponential of the coefficients.

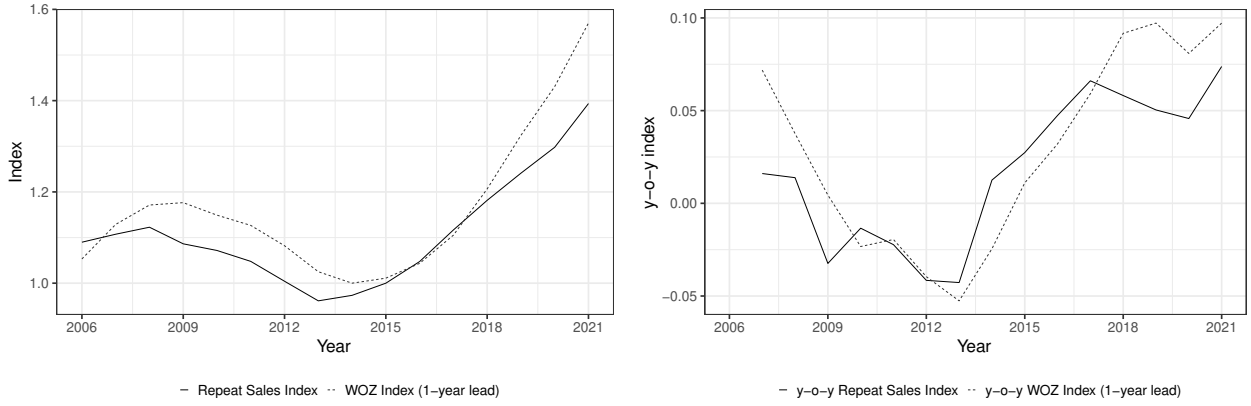
E Additional descriptive statistics

Figure E.1: Levels and year-on-year (%) change of Repeat Sales Index across postcodes



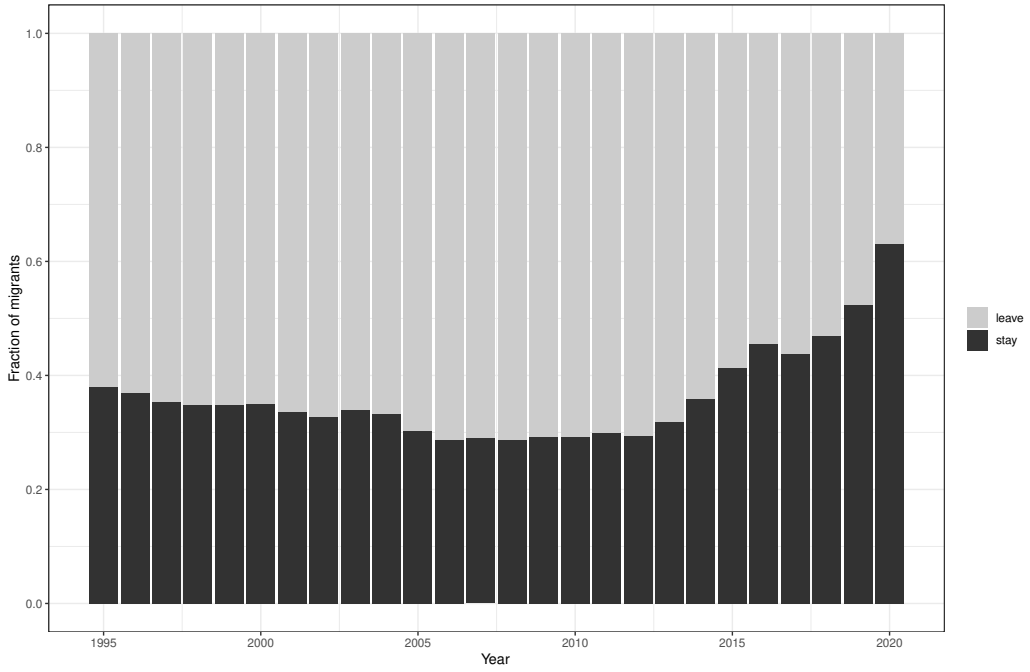
Note: Levels of repeat sales index (LHS) and year-on-year change of repeat sales index (RHS) across years and postcodes. For every year, the boxplot shows the distribution of repeat sales index values across neighborhoods. Per year, the box shows the interquartile range (25th to 75th Percentile) and the line reflects the median. The whiskers reflect the range of values within median $\pm 1.5\times$ interquartile range. Dots reflect data points outside of the whiskers. Sample of 1,795 postcodes as outlined in Table 1.
Source: *CBS Microdata*, *Kerncijfers Wijken en Buurten (KWB)*

Figure E.2: Country aggregated index of WOZ value and Repeat Sales Index



Note: Country wide index of WOZ index and repeat sales index. Both series are a weighted mean across neighborhoods, weighted by the number of dwellings. Both are index in 2015. The left-hand side shows the level of the index and the right hand side shows the year-on-year growth. The correlation between national level year-on-year WOZ and year-on-year repeat sales index is 0.86. Sample of 1,795 postcodes as outlined in Table 1.
Source: *CBS Microdata*, *Kerncijfers Wijken en Buurten (KWB)*

Figure E.3: Fraction of migrants staying / leaving by 2022



Note: Fraction of migrants still in NL by year 2022. All migrants $\geq$ 18 upon arrival. By year of first move.
Source: *CBS Microdata*

Table E.3: Descriptive statistics for migration and housing

	25th Pct.	Mean	Median	75th Pct.	N
Age at immigration	23	30,9	28	35,6	2,674,225
Number of migration spells	1	1,14	1	1	2,674,225
Duration of migration spells (days)	292	1141	653	1,431	1,400,039
Number of address spells	x	x	x	x	2,674,225
Duration at address (days)	161	727.7	381	880	5,248,119

Note: Number and moments of immigration and address spells for migrants. Duration is conditional on completing a spell.
Sample: All migrants arriving between 1995 and 2021, age $\geq$ 18 years upon arrival.
Source: *CBS Microdata*

Table E.4: Descriptive statistics for subsamples

		Sample: "100k" municipalities				
	Years	25th Pct.	Mean	Median	75th Pct.	N
Avg. WOZ value by PC4 (x 1,000)	2006 - 2021	159.63	225.58	200.08	267.09	474
Δ Avg. WOZ by PC4 (%)	2006 - 2021	-1.87	3.57	2.86	8.05	474
Housing Stock	2006 - 2021	4,010	6,226	5,552	7,929	482
Δ Housing Stock (%)	2006 - 2021	0.00	-0.02	0.00	0.00	482
Share of rental units (%)	2006 - 2021	42	56	57	70	442
Share of owner-occupied units (%)	2006 - 2021	29	43	41	58	442
Population:						
Population age ≥ 18	2006 - 2021	5,029	7,272	6,799	9,172	482
Δ Population (%)	2006 - 2021	-0.32	0.68	0.43	1.33	482
Immigrants age > 18	2006 - 2021	943	2,068	1,576	2,681	482
Δ Migrants (%)	2006 - 2021	-0.30	1.67	1.40	3.33	482
Migrant share (%)	2006 - 2021	16	27	23	35	482
Avg. household size	2006 - 2021	1.76	1.97	1.97	2.19	473
Income:						
Avg. income per worker (x 1,000)	2009 - 2021	23.8	29.5	28.1	34.0	432
Share of low-income population (%)	2009 - 2021	36.0	41.9	41.7	47.9	432
Share of high-income population (%)	2009 - 2021	12.2	19.7	18.0	25.5	430

		Sample: not "100k" municipalities				
	Years	25th Pct.	Mean	Median	75th Pct.	N
Avg. WOZ value by PC4 (x 1,000)	2006 - 2021	182.9	237.5	225.0	273.2	1,305
Δ Avg. WOZ by PC4 (%)	2006 - 2021	-2.2	2.0	1.9	5.9	1,304
Housing Stock	2006 - 2021	2,346.	4,055	3,697	5,447	1,312
Δ Housing Stock (%)	2006 - 2021	0.00	-0.02	0.00	0.00	1,312
Share of rental units (%)	2006 - 2021	27	38	35	47	1,085
Share of owner-occupied units (%)	2006 - 2021	53	62	64	72	1,085
Population:						
Population age ≥ 18	2006 - 2021	3,050	5,301	4,855	7,206	1,312
Δ Population (%)	2006 - 2021	-0.34	0.45	0.27	1.00	1,312
Immigrants age > 18	2006 - 2021	276	743	596	1,018	1,312
Δ Migrants (%)	2006 - 2021	-0.83	1.43	1.03	3.20	1,312
Migrant share (%)	2006 - 2021	8	13	12	17	1,312
Avg. household size	2006 - 2021	2.1	2.3	2.3	2.5	1307
Income:						
Avg. income per worker (x 1,000)	2009 - 2021	26.1	29.6	29.5	32.9	952
Share of low-income population (%)	2009 - 2021	36.6	40.1	40.0	43.3	953
Share of high-income population (%)	2009 - 2021	15.1	19.4	18.8	22.9	953

Note: Descriptive statistics across 4-digit postcodes and years. Δ denotes the year-on-year percentage change. Share of rental and owner-occupied units is relative to the stock of housing. Migrant share, low-income and high-income share are relative to the population on Jan 1st in a given year. Share of low-income (high-income) population is defined as the number of households that belong to the bottom 40% (top 20%) of income in the entire country. Both share of low-income and high-income population are available from the KWB dataset and only indicated if there are at least 100 households by neighborhood.
Source: *CBS Microdata*, *Kerncijfers Wijken en Buurten (KWB)*

Table E.5: Groups used for SSIV

Group	Countries
Antilles	
China	
EFTA	Liechtenstein, Norway, Switzerland
GIPS	Greece, Italy, Portugal, Spain
Germany	
India	
Indonesia	
Morocco	
Middle East	Israel, Yemen, Iraq, Iran, Afghanistan, Jordan, Pakistan, Lebanon
Middle-Eastern Europe excl. Poland	Bulgaria, Estonia, Hungary, Latvia, Lithuania, Croatia, Romania, Slovenia, Slovakia, Czech Republic
Middle-South America	
Non-EU Europe	Iceland, Serbia, North Macedonia, San Marino, Russia, Andorra, Montenegro, Moldova, Belarus, Ukraine, Bosnia and Herzegovina, Albania, Monaco, Gibraltar, Isle of Man, Kosovo, Kazakhstan, Georgia, Azerbaijan, Armenia
North Africa	Tunisia, Syria, Egypt, Algeria
North EU	Austria, Belgium, Cyprus, Denmark, France, Finland, Great Britain, Ireland, Luxemburg, Malta, Sweden
Oceania, AUS, NZ	Australia, New Zealand
Poland	
Remaining Africa	African countries except for countries in category "North Africa"
Remaining Asia	
Suriname	
Turkey	
US, Canada	

Note: We aim to construct groups of origin countries that are similar in size by using the following criteria: visa status (i.e. grouping EFTA countries), geographic proximity, language proximity. Some countries are kept as their own category dues to large migration flows (i.e. Turkey, Indonesia, Germany, Poland).
Names of countries and continents refer to geographical entities as classified by CBS.

Table E.6: Herfindahl-Hirschmann Index for migrants by country of origin, 2006 - 2022

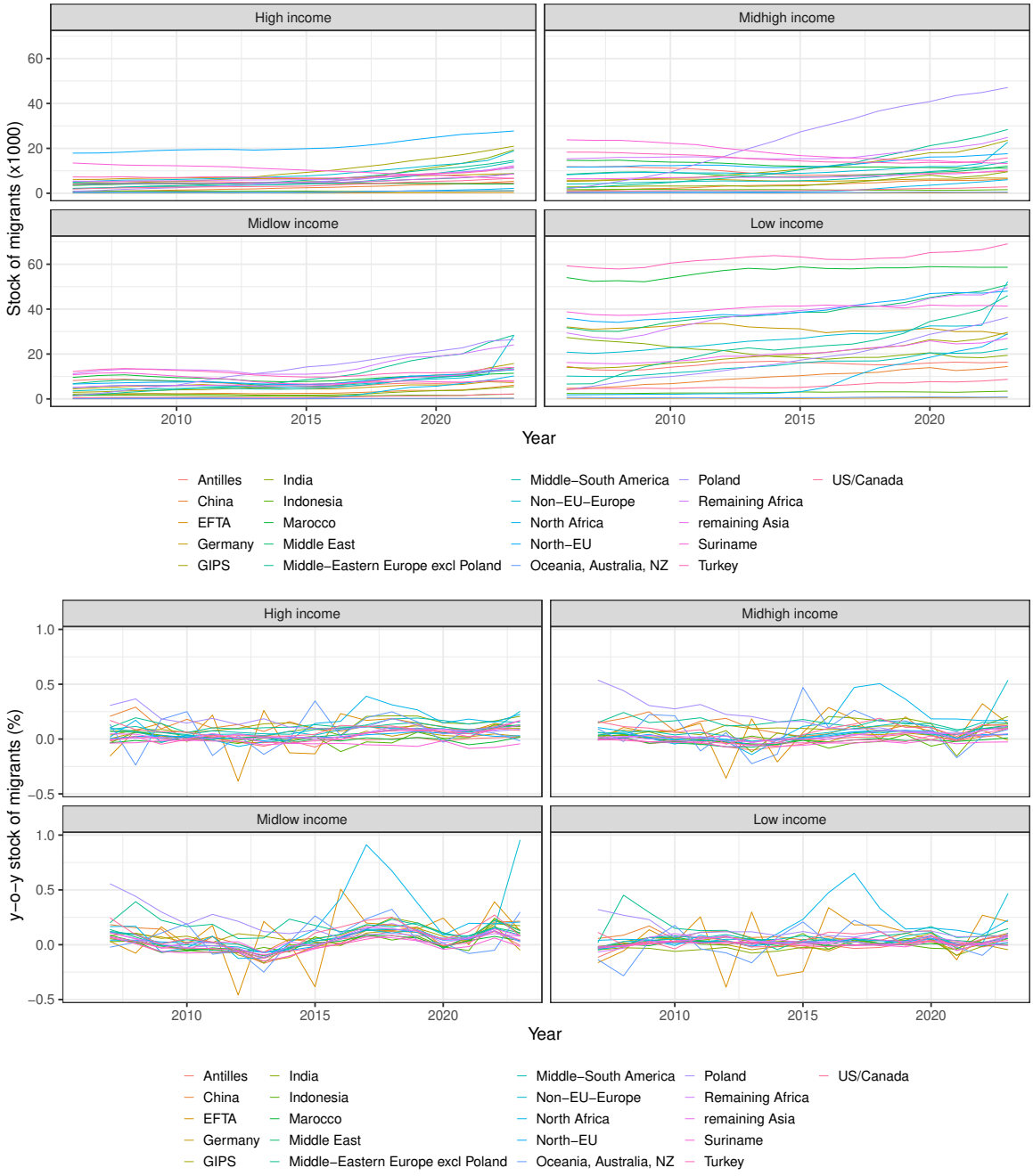
Origin group	HHI (all)	HHI (high income)	HHI (low income)	Mean (all)	SD (all)	Total inflow (all)
Antilles	0.08	0.07	0.08	73	80	665,062
China	0.13	0.14	0.17	51	60	342,583
EFTA	0.69	0.81	0.79	21	12	13,855
GIPS	0.05	0.07	0.05	64	67	902,010
Germany	0.05	0.05	0.06	59	79	879,345
India	0.08	0.14	0.04	69	85	646,985
Indonesia	0.10	0.10	0.13	24	18	124,236
Morocco	0.08	0.07	0.09	141	184	1,499,972
Middle East	0.04	0.04	0.03	68	82	949,480
Middle-Eastern Europe excl. Poland	0.05	0.06	0.06	61	53	617,345
Middle-South America	0.06	0.07	0.06	47	60	1,106,404
Non-EU Europe	0.03	0.04	0.02	52	45	997,182
North Africa	0.37	0.21	0.51	46	94	1,505,925
North EU	0.04	0.06	0.04	71	53	251,886
Oceania, Australia, NZ	0.56	0.67	0.59	29	25	30,510
Poland	0.05	0.04	0.05	65	87	1,108,929
Remaining Africa	0.06	0.05	0.06	78	120	1,387,515
Remaining Asia	0.05	0.14	0.05	53	52	737,796
Suriname	0.12	0.12	0.11	137	229	1,395,224
Turkey	0.07	0.06	0.08	149	184	1,762,391
US/Canada	0.22	0.29	0.25	50	69	218,718

Table E.7: Predictors for stock of migrants by country of origin

	Dependent variable: stock of migrants (2016 - 2022)			
	$M_{c,t}^{high}$		$M_{c,t}^{low}$	
$M_{c,t-10}^{all}$	0.078***		0.574***	
	(0.001)		(0.002)	
$M_{c,t-10}^{high}$		0.135***		
		(0.015)		
$M_{c,t-10}^{low}$				0.054***
				(0.018)
FE: 4-digit postcode	x	x	x	x
FE: Year	x	x	x	x
N.	66,391	66,391	66,391	66,391
Adj. R^2	0.428	0.336	0.887	0.284

Note: Regression of the stock of (high / low) income migrants $M_{c,i,t}^{high}$ from country origin c in postcode i and year t on the stock of migrants 10 year prior. Outcomes are measured in 2016-2022.
Sample as outlined in Section 3.
Source: *CBS Microdata*

Figure E.4: Stock and year-on-year change of the stock of migrants by region of origin and income group



Note: National-level stock of migrants by region of origin and age $\geq$ 18 upon arrival in NL. Income classes refer to the quartiles of the municipality-level income distribution.
Sample as outlined in Section 3.
Source: *CBS Microdata*

Table F.8: First stage coefficients with controls

	All migrants (1)	High income (2)	Midhigh income (3)	Midlow income (4)	Low income (5)
$\frac{\Delta M_{it}^{all}}{Pop_{it-1}}$	0.613***	0.847***	0.467***	0.861***	0.944***
	(0.201)	(0.307)	(0.111)	(0.055)	(0.084)
FE: Year	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x
Controls:	x	x	x	x	x
N	24,025	24,025	24,025	24,025	24,025
Adj. R^2	0.587	0.377	0.234	0.314	0.482
IV Wald statistic	9.3	7.6	17.8	240.6	125.7
IV F statistic	739.2	795.6	337.7	2888.2	665.3

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
First stage, as in Equation (3), with a income-type specific SSIV variable, using 10-year lagged shares. One column per income-range of migrant. Outcomes measured in 2007 - 2021. Standard errors in parentheses, clustered by municipality. The regression is weighted by the housing stock. "IV F" statistic reports the Cragg-Donald F-statistic under homoskedasticity. "IV Wald" reports the Wald statistic for weak instruments (Kleibergen-Paap rk statistic) accounting for clustered standard errors.
Source: *CBS Microdata*

F Additional results

Table F.9: OLS estimates: WOZ value

	Dependent variable: Lead of WOZ value									
	All		High income		Midhigh income		Midlow income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\frac{\Delta M_{it}^{all}}{Pop_{it-1}}$	0.479***	0.657***	1.089***	1.248***	0.180	0.537*	0.429*	0.431*	-0.023	-0.165
	(0.133)	(0.156)	(0.190)	(0.227)	(0.298)	(0.286)	(0.231)	(0.224)	(0.103)	(0.158)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: 4-digit postcode	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776	24,555	22,776
Adj. R^2	0.666	0.671	0.665	0.670	0.665	0.670	0.665	0.670	0.665	0.670

Note: First stage, as in Equation (1). One column per income-range of migrant. Outcomes measured in 2007 - 2022. Standard errors clustered by municipality.
Source: *CBS Microdata*

G External descriptive statistics

Table G.10: Excerpt from World Values Survey Wave 6 (2010 - 2014), by size of town

	Size of town (x 1,000)				
	10 - 20	20 - 50	50 - 100	100 - 500	> 500
% who *rather not* have neighbors from the following group:					
Immigrants / foreign workers	19.8	23.7	18.4	17.4	10.1
People of a different race	8	10.4	8	6.2	4.1
People who speak a different language	12.8	18.1	13.3	12.9	7.7
People of a different religion	3.7	3	1.7	2.9	2.4
% respondents who trust					
Neighborhood	74.9	71.7	71.2	64.9	50.3
People of another nationality	33.2	35.3	35.3	38.4	39.7
% respondents to agree for job priority:					
Natives over immigrants	43.9	39.2	35.5	32	24.3
Number of respondents	187	691	4141	419	169

Note: Selection of questions from Wave 6 of the World Values Survey (2010 - 2014).
For the neighborhood questions, survey participants are presented a list of sociodemographic groups and are asked to indicate all groups they would *rather not* have as neighbors. The tables show the percentage of respondents that indicate "immigrants / foreign workers" in their answer. Options are: drug addicts, people of another race, people with AIDS, migrants / foreign workers, homosexuals, people from a different religion, heavy drinkers, unmarried couples living together, people who speak a different language.
For the questions on trust, survey participants indicate their agreement on a 5 point scale from "Trust completely" to "Do not trust at all". The table above shows the percentage of respondents indicating "Trust completely" or "Trust somewhat".
For the question on job security, the table shows the percentage of respondents indicating "Agree" from the options "Agree", "Neither" and "Disagree".
Source: *World Values Survey Wave 6: 2010 - 2014*